Theory

ISPhO

A3-1

English

Frequency measurements with ring resonators

Part A. Field profile in fiber optics (2.5 points)

A1 (0.3pt)

$C_1 =$ $C_2 =$

A2 (0.5pt)

$k_{\text{in}} =$ $k_{\text{out}} =$

A3 (0.3pt)

A4 (0.3pt)

$B =$

A5 (0.3pt)

A6 (0.3pt)

A7 (0.3pt)

Equation:

A8 (0.2pt)

Equation:

Part B. Fiber resonator transmittance (3.0 points)

B1 (0.2pt)

Equation:

B2 (0.3pt)

$B_{\text{in}}(t) =$

B3 (0.3pt)

$B_{\text{in}}(t) =$

B4 (0.5pt)

$B_{\text{in}}(t) =$

B5 (0.5pt)

$N_2 =$

B6 (0.6pt)
Plot:



B7 (0.6pt)

$Q =$

Part C. Calibration of the Mach-Zehnder Interferometer (2.0 points)

C1 (0.5pt)
Plot:



C2 (1.0pt)
Plot:



C3 (0.5pt)

$\Delta\omega_{\mathrm{MZI}}/\omega_{\mathrm{MZI}} =$

Part D. Cubic nonlinearity and solitons (2.5 points)

D1 (0.1pt)

$v_g =$

D2 (0.6pt)

D3 (0.5pt)

$D_1 =$ $D_2 =$

D4 (0.3pt)

Condition on D_2:

D5 (0.6pt)
Plot:



D6 (0.2pt)

$\Delta\omega =$

D7 (0.2pt)

$\tau_s =$